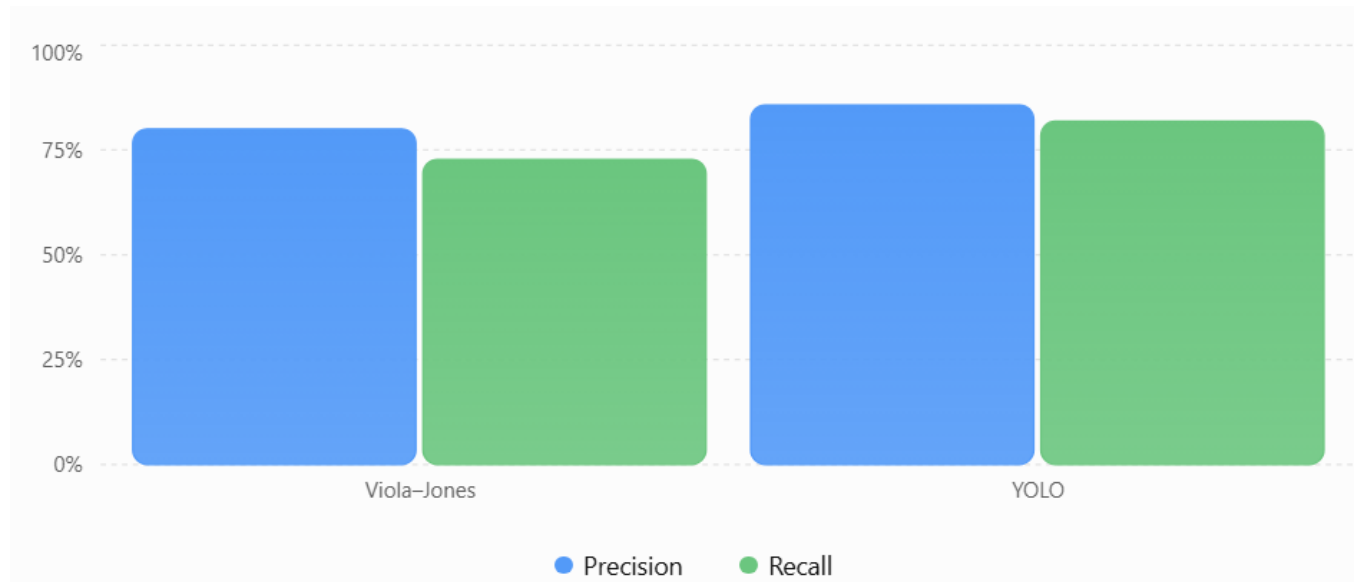


CO4_AT3 - COMPARATIVE ANALYSIS

1. Object Detection Performance Comparison

Given:

- Model A (Viola–Jones): TP = 80, FP = 20, FN = 30
- Model B (YOLO): TP = 90, FP = 15, FN = 20



Inference: YOLO has higher precision and recall than Viola–Jones. Therefore, YOLO provides better overall object detection performance.

(a) Precision and Recall

Precision:

For Model A:

Precision = 80%

Recall:

Recall = 72.73%

For Model B (YOLO):

Precision = 85.71%

Recall = 81.82%

(b) Comparison

Model	Precision	Recall
Viola–Jones	80%	72.73%
YOLO	85.71%	81.82%

(c) Conclusion

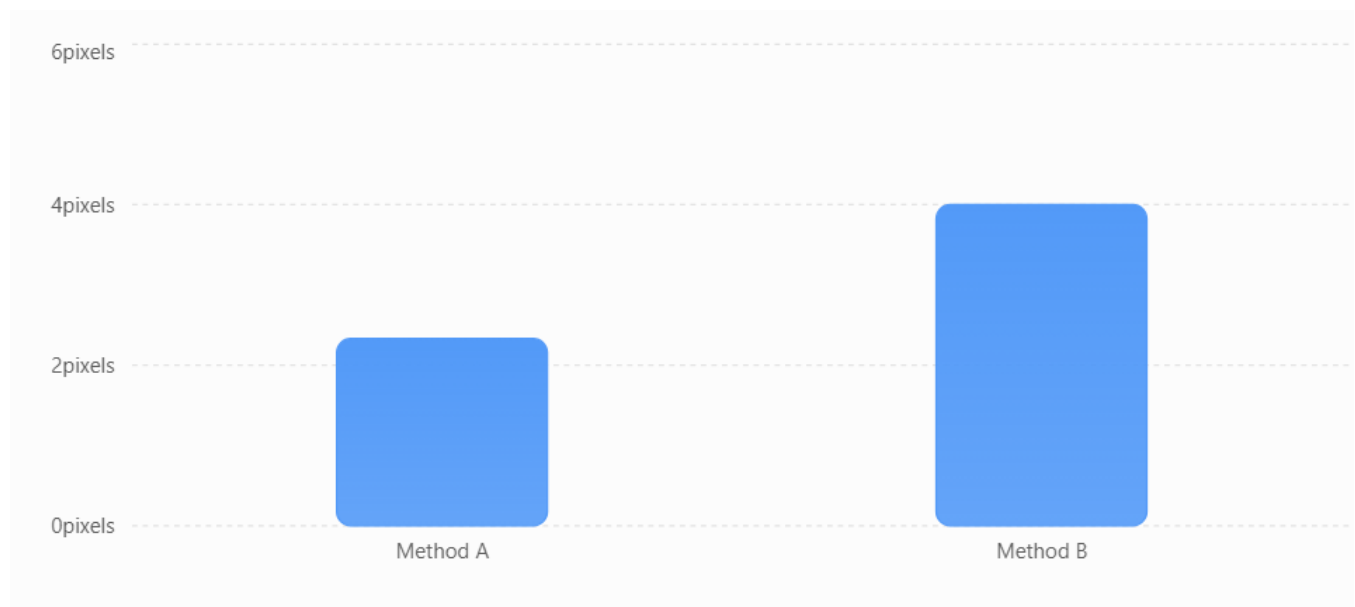
YOLO has higher precision and recall than Viola–Jones. Higher precision means fewer false detections, while higher recall means more actual objects are detected.

Therefore, YOLO provides better overall object detection performance.

2. Motion Estimation Error Analysis

Given:

- Method A: 2, 3, 2 pixels
- Method B: 3, 5, 4 pixels



(a) Average Displacement Error

The average displacement error is calculated as:

For Method A:

For Method B:

(b) Comparison

Method	Displacement Errors	Average Error
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Method A	2, 3, 2	2.33 pixels
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Method B	3, 5, 4	4 pixels
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(c) Conclusion

A lower displacement error indicates more accurate motion estimation. Method A has an average error of 2.33 pixels, while Method B has an average error of 4 pixels.

Since $2.33 < 4$, Method A has lower error and therefore provides more accurate motion estimation.

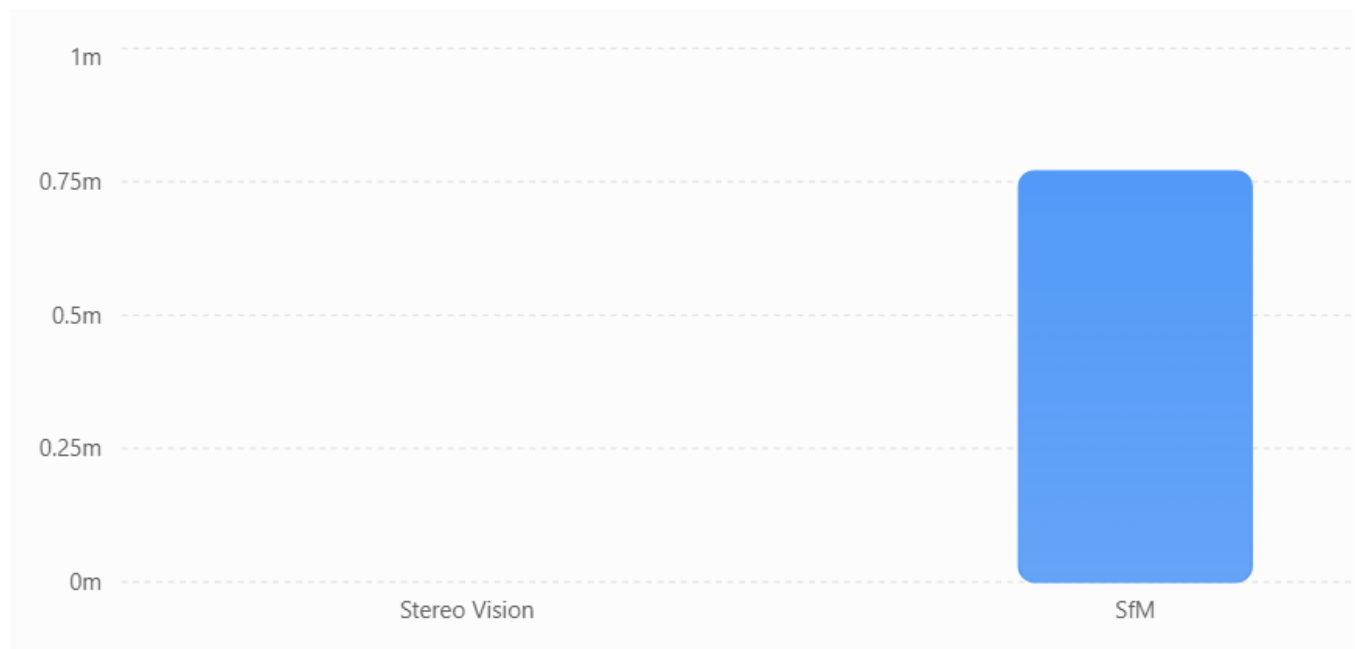
3. Depth Estimation Comparison

Given:

Actual depths = 2 m, 3 m, 4 m

Stereo Vision = 2.0 m, 3.0 m, 4.0 m

SfM = 2.5 m, 3.8 m, 5.0 m



(a) Absolute Error

The formula is:

Object 1

Stereo Vision:

SfM:

Object 2

Stereo Vision:

SfM:

Object 3

Stereo Vision:

SfM:

Error Table

Object	Actual Depth	Stereo Error	SfM Error
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1	2 m	0 m	0.5 m
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2	3 m	0 m	0.8 m
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3	4 m	0 m	1.0 m
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(b) Average Absolute Error

For Stereo Vision:

For SfM:

(c) Comparison and Conclusion

Method	Average Absolute Error
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Stereo Vision	0 m
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SfM	0.77 m
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Stereo Vision has zero average error, meaning its estimated depths exactly match the actual depths for all three objects. SfM has an average error of 0.77 m.

Since lower error indicates better accuracy:

Therefore, Stereo Vision provides more accurate depth estimation than SfM.